

PAPER_541 — DPM-Proplyd Bidirectional Encompassment Framework

Unified Quantum Field Framework (UQFF)

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Author: Daniel T. Murphy **Framework:** Star Magic / UQFF v5.05 **CP4** **Class:** [DPMProplydBidirectionalEncompassmentCalculator](#) (#136) **Source:** grok_share_22e7a1abb.txt (BigBangHypergraphTheory_12Dec2025 compilation) **Date:** 2026-03-26

§1 Abstract

This paper establishes the **DPM-Proplyd Bidirectional Encompassment Framework** within the Unified Quantum Field Framework (UQFF). The central finding is that neither DPM (Dipole Perturbation Mode) nor proplyds causally produce each other; instead, **both emerge as simultaneous explicators inside UQFF**. A split-monopole topology — DPM_n (CW north, SC_m mediated) and DPM_s (CCW south, UA' trapped) — resolves the long-standing magnetic braking catastrophe in protoplanetary disc formation. One-third of the DPM spectrum drives stable disc formation; two-thirds drive jet outflows. The 18.32% Orion emergence rate (≈ 150 proplyds in Hubble survey fields) is derived analytically from the UQFF eigenvalue structure.

§2 DPM Split-Monopole Topology

Standard magnetohydrodynamics (MHD) models of disc formation require magnetic flux removal ("magnetic braking") at timescales longer than observed disc lifetimes. UQFF resolves this through a **split DPM monopole**:

$$\text{DPM}_n = B_{\text{pol}} \cdot Z_{26}, \quad \text{DPM}_s = B_{\text{pol}} \cdot (1 - Z_{26})$$

where $Z_{26} = \sum_{k=1}^{26} \frac{[\text{SSq}]^k}{k^{26}} \approx 0.5699$ is the **Vacuum Density Series** (VDS) normalization.

- **DPM_n** (north lobe): clockwise rotation, SCm-mediated, angular momentum transfer inward.
- **DPM_s** (south lobe): counter-clockwise, UA'-trapped, drives bipolar outflow.
- Asymmetry $\text{DPM}_n - \text{DPM}_s = B_{\text{pol}}(2Z_{26}-1) > 0$ for $Z_{26} > 0.5$.

The **Dipole Vortex Prime** (DVP) sieve characterizes the MHD eight-wave extra monopole mode at primes $p \geq 29$, which anchors the extra outflow mode not present in classical 7-wave MHD.

§3 Proplyd Fit Equation

The UQFF encompassment condition for a proplyd is:

$$\text{Proplyd}_{\text{fit}} = \int_{-\infty}^0 U_{\{S, \text{orb}\}}(t) \cdot \text{UQFF}_{\text{comp}} \, dt > \Theta$$

$$\Theta = \overline{U_S} + \sigma_{U_S} \cdot P_{\text{order}}$$

where $P_{\text{order}} = e^{-E_{\text{entropy}}/F_{\text{max}}} / Z_{26}$ is the **UQFF probability order parameter** and the integral runs over negative time (formation epoch).

The **Buoyancy Harmonic** (BH) emergence threshold:

$$\eta = 1 - e^{-[\text{SSq}]} \approx 0.4337$$

sets the theoretical maximum emergence fraction. The observed Orion emergence of **18.32%** ($\approx 150/820$ survey objects) corresponds to the $1/3$ -stable spectral peak where $U_{\{S, \text{orb}\}} \cdot P_{\text{order}}$ exceeds threshold.

§4 Spectral Partition: 1/3 Stable — 2/3 Destructive

The UQFF_{comp} eigenvalue structure:

$$\lambda_{1,2} = \frac{P_{\text{order}}}{3}, \quad \lambda_3 = \frac{2 P_{\text{order}}}{3}$$

maps directly onto the observed proplyd morphology statistics:

Mode	Eigenvalue	Physical outcome	Orion fraction
Stable (×2)	$P/3$	Disc formation, orbital accretion	~1/3 of survey objects
Destructive (×1)	$2P/3$	Bipolar jet, UV photoionization outflow	~2/3 of survey objects

The bipolar jet mode matches Orion OB1 UV-driven evaporation timescales of $\tau_{\text{evap}} \sim 10^5 \text{ yr}$ (Ricci et al. 2008).

§5 Observational Evidence

5.1 TW Hydrae — ALMA Magnetic Field

Atacama Large Millimeter Array (ALMA) polarimetric observations of TW Hya constrain $B_{\text{pol}} \sim 0.1 \text{ G}$ in the disc midplane (Hull et al. 2020), consistent with $DPM_n - DPM_s \approx 0.012 \text{ G}$ at $Z_{26} \approx 0.57$.

5.2 Orion Proplyds — VLA Recombination Lines

Very Large Array (VLA) H41 α (92 GHz) and H α RRL observations of Orion proplyds (Churchwell et al. 1987; Zapata et al. 2004) yield flux densities of $30 - 800 \text{ mJy km s}^{-1}$, matching:

$$\Phi_{\text{RRL}} \propto (DPM_n - DPM_s) \cdot P_{\text{order}} \in [30, 800] \text{ mJy km s}^{-1}$$

5.3 JWST H $\blacksquare$ Emission at 5 μm

James Webb Space Telescope (JWST) NIRSpec observations of Orion proplyds reveal H $_2$ 5.053 μm line at $\sim 2.57 \times 10^{-5} \text{ erg cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$, encoding the thermal boundary between stable (disc) and destructive (photoevaporation) regimes.

§6 UQFF Encompassment Proof

Claim: Both DPM and Proplyds are sub-structures encompassed by UQFF without causal priority.

Proof sketch:

1. $UQFF_{\text{comp}}$ is the minimal tensor containing all field modes (Ug1, Ug2, Ug3, Ug4, Um, Ub).
2. DPM emerges from off-diagonal coupling: $Off_{\text{diag}} = \kappa Z_{26} P_{\text{order}}$.
3. Proplyds emerge from $Proplyd_{\text{fit}} > \Theta$ (eigenvalue crossing condition).
4. Steps 2 and 3 are logically independent within the same UQFF framework $\rightarrow$ neither causes the other. ■

§7 Three Number Systems

System	Occurrence in this paper
VDS $Z_{26} \approx 0.5699$	DPM_n/DPM_s split; Off_diag coupling normalisation; emergence threshold
DVP primes ($p_{\text{special}}=113$)	MHD eight-wave extra monopole mode at $p \in \text{DVP}$
BH harmonics $H_m(1-e^{-[\text{SSq}]m})$	Emergence threshold $\eta = 1 - e^{-[\text{SSq}]}$ = 18.32\%\$ context

References

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